NumPy Cheat Sheet with Examples and Outputs

1. Array Creation Functions

Function	Description	Example	Output
np.array()	Create an array	np.array([1,2,3])	array([1, 2, 3])
np.zeros()	All zeros	np.zeros((2,2))	[[0., 0.],[0., 0.]]
np.ones()	All ones	np.ones((2,3))	[[1.,1.,1.],[1.,1.,1.]]
np.arange()	Range of numbers	np.arange(0,10,2)	[0,2,4,6,8]
np.linspace()	Evenly spaced	np.linspace(0,1,5)	[0.,0.25,0.5,0.75,1.]

2. Array Inspection & Attributes

Function	Description	Example	Output
arr.shape	Dimensions	np.array([[1,2,3],[4,5,6]]).shape	(2,3)
arr.ndim	Number of dimensions	arr.ndim	2
arr.size	Total elements	arr.size	6

3. Arithmetic & Math Operations

Function	Description	Example	Output
np.add()	Addition	np.add([1,2],[3,4])	[4,6]
np.subtract()	Subtraction	np.subtract([5,4],[2,1])	[3,3]
np.multiply()	Multiplication	np.multiply([1,2],[3,4])	[3,8]
np.divide()	Division	np.divide([6,8],[2,4])	[3.,2.]
np.sqrt()	Square root	np.sqrt([1,4,9])	[1.,2.,3.]

4. Statistical Functions

Function	Description	Example	Output
np.mean()	Mean	np.mean([1,2,3])	2.0
np.median()	Median	np.median([1,3,2])	2.0
np.std()	Standard deviation	np.std([1,2,3])	0.816...
np.sum()	Sum	np.sum([1,2,3])	6
np.max()	Maximum	np.max([1,2,3])	3

5. Linear Algebra Functions

Function	Description	Example	Output
np.dot()	Dot product	np.dot([1,2],[3,4])	11
np.linalg.inv()	Inverse matrix	np.linalg.inv([[1,2],[3,4]])	[[-2.,1.],[1.5,-0.5]]
np.linalg.det()	Determinant	np.linalg.det([[1,2],[3,4]])	-2.0

6. Random Functions

Function	Description	Example	Output
np.random.rand()	Uniform random [0,1)	np.random.rand(2,2)	Random values
np.random.randint()	Random integers	np.random.randint(0,10,(2,3))	Random int matrix
np.random.seed()	Set seed	np.random.seed(0)	Reproducible results